

Philosophy of the Beat: Bergson, Heidegger and Local Time

Benjamin Brécheteau

March 1, 2026

Standalone article extending the treatise *Time May Not Exist!* (Brécheteau, 2025)

DOI: 10.5281/zenodo.17214502

brecheteaub@gmail.com

Abstract

This article proposes an *operational* articulation between Bergson’s *duration*, Heidegger’s *temporality*, and a minimal physical model: the Chronon Field $\Phi(x)$. Rather than a universal flow or a mere coordinate, time is treated here as a *local beat* of coherence. We show how this hypothesis fits within the existing experimental scaffolding (definition of the second by caesium, NTP/PTP synchronization, gravitational redshift, Lamb shift, Casimir), and we suggest *differential* tests (co-located qubits at $\Delta h \sim 0.3$ m, clock comparisons at 10^{-18}) to assess the idea that a *compatibility of rhythms*, more than a global simultaneity, underpins *public time*. *Operational clarification*: by *beat* we mean the local coherence period $T(x) = 1/\Phi(x)$ (with $\omega(x) = 2\pi\Phi(x)$); by *cycle*, the interval of *holding* within which a phenomenon remains compatible with its environment. The window metric $\tau_{\text{win}} \sim 1/\Delta\Phi$ denotes the inverse of the frequency mismatch (in Hz) between local rhythms. Phenomenology of openness thereby gains a metric of holding; physics gains ontological depth without abandoning covariance.

Keywords: Chronon Field, duration, local synchronization, public time, beat, general relativity, phenomenology, time metrology

Philosophy of the Beat: Bergson, Heidegger and Local Time

I — The misunderstanding of duration

For more than a century, two images of time have coexisted without really speaking to each other. On one side, physics unfurls a measurable, parameterized time, cut into abstract units. On the other, philosophy speaks of lived duration, of presence, of an inner flow that escapes any measurement. Between the clocks’ second and the heart’s instant, the divorce seemed complete.

Henri Bergson called pure duration this continuous experience that is not counted but felt: a passage without juxtaposition, where becoming precedes form. Martin Heidegger, for his part, saw in temporality the very essence of being: not a framework, but an opening of the world by Dasein, this “standing-in-time” that makes any presence possible.

Science, for its part, thought it had to choose: either duration is subjective, or time is geometric. But the Chronon Field, $\Phi(x)$, proposes a third path: a rhythmic background where these two temporalities meet. This proposal does not abolish measurement; it relocates it within a skeleton of synchronization that makes a “public time” possible without denying the locality of lived experience.

Example of this tension: the international second is defined by the hyperfine transition of caesium at 9,192,631,770 Hz, while a computer network synchronized by NTP aligns to a few 10^{-3} s, and a datacenter using PTP (IEEE 1588) with hardware timestamping holds $< 10^{-6}$ s, even $\sim 10^{-7}$ s locally. Between these layers of precision slips lived experience, whose perceptual resolution does not align with the protocols’ metric but with a beat of attention and action.

Conventions (operational). We adopt the *dimensional* convention $[\Phi] = \text{s}^{-1}$, with the practical equivalence $\tilde{\Phi} \equiv \Phi/\Phi_0$ (dimensionless) when needed: the predictions are invariant. Operationally,

$$d\tau = \Phi^{-1}(x) dt.$$

In the weak-field regime, we can use the *phenomenological dictionary* $\mathbf{g} \simeq -c^2 \nabla \ln \Phi$ to speak of temporal gradients — without modifying light cones nor introducing new energy. Canonical definitions are recalled in Box .

Box — Operational toolbox.

$$\boxed{\omega(x) = 2\pi \Phi(x), \quad T(x) = \Phi(x)^{-1}, \quad [\Phi] = \text{s}^{-1}} \quad (\text{BO})$$

- **Metrological reading:** compare *dimensioned residuals* (after GR subtraction); favor differential protocols (co-location, permutations, long baselines).
- **Frame invariants:** GR geometry unchanged; no $T_{\mu\nu}$ added; Φ reparametrizes durations, carries no energy.
- **Limits of the analogy:** “beat”, “breathing”, “cycle” *designate* $T(x)$ and windows of compatibility; *no* absolute periodicity is postulated. Technical sections: *no metaphors*.

Box — Link to Article 9 (neuro/cognition), methodological beacons.

- **Demonstrable/measurable (here):** clock residuals, NTP/PTP/GNSS synchronization, weak-field dictionary.
- **Controlled hypothesis (Art. 9):** correspondences with perceptual windows and cortical rhythms; to be tested *without* introducing any additional “neuronal” Φ .

II — Pure duration revisited

In Bergson, duration is neither a succession of instants nor a spatialized continuum. It is tension and relaxation where the past prolongs itself into the present without closing in it. Within the Chronon Field framework, this intuition becomes *operational*: each local event is indexed to a *holding time* (see Box) that conditions the compatibility of phenomena.

Duration thus becomes *localized*: not a universal flow, but a plurality of rhythms composing a fabric of coherences. The present does not “flow”; it is *recomposed* through windows $T(x) = 1/\Phi(x)$ within which a phenomenon can be maintained.

Operational test (compatibility window). If holding depends on an agreement of rhythms, the coherence window must shrink with frequency mismatch. We model, at first order,

$$\tau_{\text{win}} \sim \frac{1}{|\Delta\Phi|},$$

where $\Delta\Phi$ is the gap (in Hz) between the phenomenon’s own cadence and that of its environment, or the gap between two experimental sites. *Bounding by losses*: if the coherence amplitude C obeys $\dot{C} = (i\Phi - \Gamma)C$ (phase/loss split), then $\tau_{\text{win}} \lesssim \min(|\Delta\Phi|^{-1}, \Gamma^{-1})$.

Experimental landmarks (measurable here).

- **Ramsey metrology.** Ramsey-type interrogations bound the coherence of an atomic oscillator by a duration T that operationally realizes the window τ_{win} ; contrast decays as soon as $|\Delta\Phi|T \gtrsim 1$.
- **Correlated clocks.** In fibre/GNSS comparisons between sites, a differential phase drift $\Delta\phi(t) \propto \int \Delta\Phi dt$ directly constrains $|\Delta\Phi|$; 10^{-18} stability opens access to the centimeter equivalent (via the weak-field dictionary).
- **Avalanche detectors.** Instrumental dead time ($\sim 10\text{--}100$ ns) bounds detection cadence: an explicit minimal granularity of access, independent of any metaphor.

Controlled bridge to Art. 9 (neuro/cognition).

- **Hypothesis:** flicker fusion (critical flicker fusion, CFF) around 50–60 Hz as a *psychophysical aggregation marker* could illustrate a constraint of rhythmic compatibility.
- **Not a proof:** these observations do not prove $\Phi(x)$; they provide *calibration landmarks* to be tested without postulating a distinct “neuronal” field.

Thus framed, Bergsonian duration is neither annulled nor mystified: it acquires a local metric ($T, \Delta\Phi, \Gamma$) and readout procedures (Ramsey, correlated comparisons) that allow it to be *falsified* if needed.

III — Being and beat

Heidegger does not describe time as a backdrop, but as an *opening*: Dasein does not simply “find itself” in time, it *is* temporal. Within the Chronon Field framework, $\Phi(x)$ does not provide an external clock; it *makes a local present possible* by giving a scale of compatibility (see Box) at which phenomena hold or dissolve. Openness is thus not an undifferentiated flow: it is a *local condition of holding*, testable.

Operational parallel (without metaphor). Any measurement proceeds in three phases: preparation, interrogation, readout. It implements a *readout window* τ_{gate} (gate time) that bounds the information extracted. Two simple consequences:

- **C1 (instrumented present).** The published observable is a weighted average over τ_{gate} ; experimental “simultaneity” is *defined* by this choice.
- **C2 (cadence vs visibility).** For resonant protocols (e.g. Ramsey), visibility/contrast decays when $|\Delta\Phi|\tau_{\text{gate}} \gtrsim 1$; one then reads a coherence dependence governed by $T(x) = 1/\Phi(x)$ without assuming absolute periodicity.

Falsifiable (local) prediction: at fixed Γ , varying τ_{gate} and $\Delta\Phi$ must shift the loss-of-visibility threshold in a way compatible with $\tau_{\text{win}} \lesssim \min(|\Delta\Phi|^{-1}, \Gamma^{-1})$. Systematic failure $\Rightarrow$ upper bound on the effect of Φ in the tested regime.

Experimental landmarks (controllable).

- **Optical clocks.** Fringe contrast (interrogation of duration T) drops when $|\Delta\Phi|T \gtrsim 1$; control loops in any case impose an effective window (T_{cycle}): one can sweep T and Δh to isolate the expected dependence.
- **Long-baseline comparisons.** Between sites A, B , phase drift $\Delta\phi(t) \propto \int_0^t \Delta\Phi dt'$ gives direct access to $|\Delta\Phi|$; 10^{-18} stability allows a centimeter equivalent via the weak-field dictionary ($\nabla \ln \Phi \simeq \nabla \psi/c^2$).
- **Loss discriminants.** With $\dot{C} = (i\Phi - \Gamma)C$, the phase/loss split requires independent witnesses of Γ (dynamical decoupling, T_1/T_ϕ control) to avoid confusing rhythmic mismatch with environmental noise.

Box — Limits of the analogy (Heidegger $\leftrightarrow \Phi$).

- Heideggerian “openness” guides the *sense* of the tests; it introduces no new physical assumption.
- No universal periodicity is postulated: $T(x)$ is a *local scale of compatibility*, not a cosmic metronome.
- GR geometry unchanged; no $T_{\mu\nu}$ added; all results are read *differentially*.

Link to Art. 9 (neuro/cognition) — beacons.

- **Hypothesis:** some attentional performances (windows 30–300 ms) could optimize synchronous readout in networks of clocks/qubits under the same environment.
- **Measurable:** correlate synchronization metrics (Allan variance, visibility) with attentional tasks under controlled *jitter*; postulate *no* distinct “neuronal” Φ .

IV — The local against the absolute

Classical physics aimed at a homogeneous time; phenomenology insisted on situation. With the Chronon Field, these views converge: time is *local*, yet *metric* and *comparative*. Coherence arises from a *compatibility*

of *rhythms* between regions, without postulating an absolute.

GR constraints (measurable). Gravitational redshift imposes

$$\frac{\Delta\nu}{\nu} \simeq \frac{\Delta U}{c^2} \simeq \frac{g \Delta h}{c^2} \approx 1.1 \times 10^{-16} \text{ m}^{-1} \times \Delta h.$$

Optical clocks at 10^{-18} resolve $\Delta h \sim \text{cm}$ (fibre/GNSS comparisons). Reference history (order of magnitude): Pound–Rebka (Mössbauer, 1960), Hafele–Keating (aircraft, 1971), then modern optical links ($\sim 10^{-18}$).

Weak-field dictionary (anchoring). We adopt

$$\nabla \ln \Phi \simeq \frac{\nabla \psi}{c^2}, \quad \Delta \ln \Phi \simeq \frac{\Delta \psi}{c^2},$$

where ψ is the Newtonian potential. Thus $\nabla \Phi(x)$ carries the imprint of the potential: compatibility of rhythms conditions the holding of “public time” *without* modifying light cones nor introducing energy (no $T_{\mu\nu}$).

Technical skeleton (synchronization/public).

- **GNSS:** disciplining of local oscillators, typical dispersion $\sim$ tens of ns (wide-area network).
- **PTP (IEEE 1588):** hardware timestamping, $\lesssim 10^{-6}$ s at datacenter level (down to $\sim 10^{-7}$ s locally).
- **NTP:** office/general networks, $\sim$ ms.

Three layers of constraint; the “public” does not erase the local, it orchestrates it via measurable compatibility criteria.

Procedures and null-tests (differential readout).

- **Pipeline:** co-location $\rightarrow$ controlled height step $\Delta h \rightarrow$ independent long baselines (fibre/GNSS) $\rightarrow$ link permutations $\rightarrow$ inter-site correlations.
- **Dimensioned residual:** definition and label for later cross-reference

$$\left(\frac{\Delta\nu}{\nu}\right)_{\text{res}} = \left(\frac{\Delta\nu}{\nu}\right)_{\text{obs}} - \left(\frac{\Delta\nu}{\nu}\right)_{\text{GR}} = \varepsilon_{\Phi}. \quad (1)$$

- **Acceptance criteria:** link independence; dimensionless witnesses neutral; persistence under permutations; bound $|\varepsilon_{\Phi}| < 10^{-18}$ in the absence of effect.

Link to Art. 9 (neuro/cognition) — beacons.

- **Hypothesis:** attentional windows 30–300 ms as *operational* compatibility constraints for situated action.
- **Demonstrable here:** no postulate of a “neuronal” field; correlate only public synchronization metrics (Allan variance, visibility) with tasks under controlled *jitter*.

Box — Limits of the analogy (technical section).

- The terms “beat/rhythm” *designate* $T(x) = 1/\Phi(x)$ and windows of compatibility; no absolute periodicity is postulated.
- Reading strictly differential; GR geometry unchanged; no new $T_{\mu\nu}$.

V — Conclusion: operational duration

Between Bergson and Heidegger, a clear point emerges: time is neither a backdrop nor a flow, but a *beat of openness*. The Chronon Field makes this intuition *operational*: it gives duration a local physical basis (see Box) and is stated without modifying GR geometry nor introducing new $T_{\mu\nu}$: everything is read *differentially*.

Two “vacuum” landmarks recall a non-trivial background: the Lamb shift (1947) reveals a correction of order ~ 1.06 GHz between $2S_{1/2}$ and $2P_{1/2}$ of hydrogen; Lamoreaux’s experiment (1997) measures the Casimir force at the micronewton level for submicrometric gaps. These facts do not prove $\Phi(x)$; they indicate that a structuring background is at work. The Chronon Field fits as a *minimal hypothesis*: not additional energy, but a *rhythmicity of coherence*.

Empirical program (targeted falsifiability).

- **Differential clocks.** Two optical clocks co-located then separated by $\Delta h \sim 0.3$ m; expected drift $\Delta\nu/\nu \simeq (g/c^2) \Delta h \approx 1.1 \times 10^{-16} \text{ m}^{-1} \times \Delta h$, extrapolable to the centimeter ($\sim 10^{-18}$) at 10^{-18} stability; objective: link the drift to $\nabla \ln \Phi(x)$ via the weak-field dictionary.
- **Qubits in a gradient.** Two identical qubits (cQED/ions) at distinct potentials (Δh); sequences synchronized to clocks to test a window law $\tau_{\text{win}} \sim 1/|\Delta\Phi|$ at *controlled losses*. Loss model: $\Gamma = \Gamma_{\text{env}} + \xi \Phi + b|\nabla\Phi|$; disentangle ξ, b via T_1/T_ϕ witnesses and decorrelations (dynamical decoupling).
- **Long-baseline Ramsey.** Sites A, B linked by fibre/GNSS; read $\Delta\phi(t) \propto \int \Delta\Phi dt$ and constrain $|\Delta\Phi|$ directly; link permutations and null-tests to exclude transport biases.

Language clarification (metaphor $\rightarrow$ model). The words “cycle”, “breathing”, “beat” *strictly designate* $T(x) = 1/\Phi(x)$ and the compatibility windows τ_{win} . No absolute periodicity is postulated. In the technical sections, metaphors are banned in favor of the quantities $(T, \Delta\Phi, \Gamma)$ and the protocols (Ramsey, correlated comparisons).

Box — Controlled bridge to Art. 9 (neuro/cognition).

- **Demonstrable here:** public synchronization metrics (Allan variance, visibility/contrast), differential residuals, controlled dependence on $|\Delta\Phi|$ and Γ .
- **Hypothesis (Art. 9):** correspondences *without postulating a distinct neuronal field* with attentional windows (30–300 ms) and psychophysical thresholds (CFF ~ 50 –60 Hz) — to be tested via instrumented correlations and controlled *jitter*.

Consolidated null-test (multi-site campaign).

$$\max_{\text{links, perms, sites}} |\varepsilon_\Phi| < 10^{-18} \quad (95\% \text{ CL}) \Rightarrow \text{otherwise: reject the signal attributed to } \Phi.$$

Duration is no longer a mystery of the mind nor time an empty abstraction. With $\Phi(x)$, they meet in a measurable *rhythm of being*: the real does not “flow”, it *holds* and recomposes itself at each window $T(x)$. *Series cross-references.* For the intuition of “poly-rhythmic world” openness, see Article 1, *The Time That Beats*. For the thermodynamic formulation (entropic gradient, weak-field dictionary), see Article 8, *Entropic Gravity and Temporal Gradients*. The consolidated experimental agenda is given in Article 12, *The Year of Chronon*.

References

- [1] B. Brécheteau, *Le temps n'existe peut-être pas !*, treatise on the Chronon Field, 2025 (Zenodo preprint).
- [2] B. Brécheteau, “Le Temps qui bat : redéfinir la pulsation du réel”, standalone article extending the treatise, 2025 (Zenodo preprint).
- [3] B. Brécheteau, “Chronon Field and the End of Timeless Physics”, standalone article extending the treatise, 2025 (Zenodo preprint).
- [4] B. Brécheteau, “Entropic Gravity and Temporal Gradients: $\Phi(x)$ as a Thermodynamic Vector”, Article 8 of the Chronon Field series, 2026 (Zenodo preprint).
- [5] B. Brécheteau, “The Year of Chronon”, consolidated experimental agenda of the series, 2026 (in preparation).
- [6] H. Bergson, *Essai sur les données immédiates de la conscience*, Alcan, Paris, 1889.
- [7] H. Bergson, *Durée et simultanéité. À propos de la théorie d'Einstein*, Alcan, Paris, 1922.
- [8] M. Heidegger, *Sein und Zeit*, Niemeyer, Tübingen, 1927 (Engl. transl. *Being and Time*, Blackwell, 1962).
- [9] A. Einstein, “Zur Elektrodynamik bewegter Körper”, *Annalen der Physik* **17**, 891–921 (1905).
- [10] R. V. Pound, G. A. Rebka Jr., “Apparent weight of photons”, *Phys. Rev. Lett.* **4**, 337–341 (1960).
- [11] J. C. Hafele, R. E. Keating, “Around-the-world atomic clocks: Predicted relativistic time gains”, *Science* **177**, 166–168 (1972).
- [12] N. Ashby, “Relativity in the Global Positioning System”, *Living Rev. Relativ.* **6**, 1 (2003).
- [13] A. D. Ludlow, M. M. Boyd, J. Ye, E. Peik, P. O. Schmidt, “Optical atomic clocks”, *Rev. Mod. Phys.* **87**, 637–701 (2015).
- [14] W. F. McGrew *et al.*, “Atomic clock performance enabling geodesy below the centimetre level”, *Nature* **564**, 87–90 (2018).
- [15] J. Grotti *et al.*, “Geodesy and metrology with a transportable optical clock”, *Nature Phys.* **14**, 437–441 (2018).
- [16] D. W. Allan, “Statistics of atomic frequency standards”, *Proc. IEEE* **54**, 221–230 (1966).
- [17] F. Riehle, *Frequency Standards: Basics and Applications*, Wiley-VCH, 2004.
- [18] D. Mills *et al.*, “Network Time Protocol Version 4: Protocol and Algorithms Specification”, RFC 5905, IETF (2010).
- [19] IEEE Instrumentation and Measurement Society, *IEEE Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems (IEEE 1588)*, IEEE Std 1588-2008 and 1588-2019.
- [20] W. E. Lamb Jr., R. C. Retherford, “Fine Structure of the Hydrogen Atom by a Microwave Method”, *Phys. Rev.* **72**, 241–243 (1947).
- [21] H. B. G. Casimir, “On the attraction between two perfectly conducting plates”, *Proc. K. Ned. Akad. Wet.* **51**, 793–795 (1948).
- [22] S. K. Lamoreaux, “Demonstration of the Casimir Force in the 0.6 to 6 μm Range”, *Phys. Rev. Lett.* **78**, 5–8 (1997).
- [23] D. H. Kelly, “Visual responses to time-dependent stimuli. I. Amplitude sensitivity measurements”, *J. Opt. Soc. Am.* **51**, 422–429 (1961).
- [24] S. Dehaene, *Consciousness and the Brain*, Viking, New York, 2014 (perceptual temporal windows and integration).

